159102 workshop – Week 9 – Some selected solutions

4. What colour do you get if the three RGB values are the same? E.g. 150, 150, 150. Leave out the small numbers because we know that 0, 0, 0 is black. And leave out the large numbers because we know that 255, 255, 255 is white. Look somewhere in between. Note that this colour can appear in many shades.

Solution to Q4:

The colour is grey (also spelled gray). There are many shades of grey.

5. Assume you want to write a program to create an image in a file called smallimage.ppm. Your image is going to be 400 pixels down the screen and 250 pixels across the screen. Write down the first four lines that will appear in the file called smallimage.ppm.

Solution to Q5:

```
P3
# smallimage.ppm
250 400
256
```

7. What does the following system command do: F:
(*if you are unable to do this on your computer, don't worry about it*)

Solution to Q7:

This system command changes the drive to drive F: This only works if you have a drive F: (e.g. a usb drive). You could enter C: to change to the main hard drive on your computer.

8. The following program is meant to create the image of a green rectangle. However there are five (5) errors in the program. Identify each error and state what needs to be done to correct it.

```
#include <iostream>
#include <fstream>
using namespace std;

class pixel_class {
private:
  int red, green, blue;
public:
  void loaddata(int v1, int v2, int v3);
  void datatofile(fstream ppmfile);
  int getR() { return red; }
  int getG() { return green; }
  int getB() { return blue; }
};

void createpicture();
```

```
pixel_class picture[500][300];

int main() {
  int x, y;
  fstream myfile;
  createpicture();
  filename = "myImage.ppm";
  myfile.open(filename.c_str(), fstream::out);
  myfile << "# " << filename << endl;
  myfile << 250 << " " << 100 << endl;
  myfile << 256 << endl;
  for (y = 0; y < 300; y++) {
    for (x = 0; x < 500; x++) {
      picture[x][y].datatofile(myfile);
    }
    myfile << endl;
  }
  myfile.close();
}

void createpicture() {
  int x, y;
  for (y = 0; y < 300; y++) {
    for (x = 0; x < 500; x++) {
      picture[x][y].loaddata(255, 153, 0);
    }
  }
}

//--------------- methods for the pixel_class ------------
void pixel_class::loaddata(int v1, int v2, int v3) {
  red = v1;
  green = v2;
  blue = v3;
}

void pixel_class::datatofile(fstream & ppmfile) {
  // write the data for one pixel to the ppm file
  ppmfile << red << " " << green;
  ppmfile << " " << blue << "  ";
}
```

Solution to Q8:

1. The function prototype of the method datatofile must have a call by reference parameter. Change to:
`void datatofile(fstream & ppmfile);`

2. The variable filename is not defined. Include the line: `string filename;`

3. The first line of the header information for the .ppm file is missing. Include the line:
`myfile << "P3\n";`

4. The size information in the header for the .ppm file is not correct. Change it to:

```
myfile << 500 << " " << 300 << endl;
```

5. The rectangle should be green. The incorrect RGB values are loaded. Change to:
```
picture[x][y].loaddata(0, 255, 0);
```
Note: other shades of green are possible.